

Using Φ to denote the set of units to be forced, we will let d_i be the trajectory that we will force y_i to follow, for each $i \in \Phi$. So we set

$$f_i = \frac{dd_i}{dt} - F_i \quad (3.54)$$

and $y_i(t_0) = d_i(t_0)$ for $i \in \Phi$ and $f_i = 0$ for $i \notin \Phi$, with the consequence that $y_i = d_i$ for $i \in \Phi$. Now let the error functional be of the form

$$E = \int_{t_0}^{t_1} L(f(t), t) dt, \quad (3.55)$$

where typically $L = \sum_{i \in \Phi} f_i^2$.

We can modify the derivation in section 3.3.1 for this teacher forced system. For $i \in \Phi$ a change to $\tilde{y}_i$ will be canceled immediately, so taking the limit as $\Delta t \rightarrow 0$ yields $z_i = 0$. Because of this, it doesn't matter what e_i is for $i \in \Phi$.

We can apply (3.18) to calculate e_i for $i \notin \Phi$. The chain rule is used to calculate how a change in y_i effects E through the f_i , yielding

$$e_i = \sum_{j \in \Phi} \frac{\delta E}{\delta f_j} \frac{\partial f_j}{\partial y_i}$$

or

$$e_i = \sum_{j \in \Phi} \frac{\partial L}{\partial f_j} - \frac{1}{T_j} \sigma'(x_j) w_{ij} \quad (3.56)$$

For $i \notin \Phi$ (3.26) and (3.37) are unchanged, and for $j \notin \Phi$ and any i (3.27) also remains unchanged. The only equations still required are $\partial E / \partial w_{ij}$ for $j \in \Phi$ and $\partial E / \partial T_i$ for $i \in \Phi$. To derive the first, consider the instantaneous effect of a small change to w_{ij} , giving

$$\frac{\partial E}{\partial w_{ij}} = \frac{1}{T_j} \int_{t_0}^{t_1} y_i \sigma'(x_j) \frac{\partial L}{\partial f_i} dt. \quad (3.57)$$

Analogously, for $i \in \Phi$

$$\frac{\partial E}{\partial T_i} = -\frac{1}{T_i} \int_{t_0}^{t_1} \frac{\partial L}{\partial f_i} \frac{dy_i}{dt} dt. \quad (3.58)$$

We are left with a system with a number of special cases depending on whether units are in Φ or not. Interestingly, an equivalent system results if we leave (3.26), (3.27), and (3.37) unchanged except for setting $z_i = \partial L / \partial f_i$ for $i \in \Phi$ and setting all the $e_i = 0$. It is an open question whether there is some other way of defining z_i and e_i that results in this simplification.

However, by taking the limit as the step size goes to zero, it is possible to show that the continuous time analogue of teacher forcing is to force the output states to follow desired trajectories, with the error being the difference between the derivative that the network attempts to apply to these units and the derivative of the desired